

Tutorial 2

A mobile Service provider wants to Classify SMS Messages as Spam or not Spam Using a Single Layer Neural Network with 5 hidden neurons in the layer.

x_1	x_2	x_3	Target.
4	2	1	1
1	0	1	0
6	3	1	1
2	1	1	0

w_1, w_2, b

$$w_1 \quad 0.1 \quad -0.2 \quad 1$$

$$w_2 \quad 0.3 \quad -0.1 \quad 1$$

$$w_3 \quad 0.2 \quad 0.4 \quad 1$$

$$w_4 \quad 0.1 \quad 0.3 \quad 1$$

$$w_5 \quad -0.2 \quad 0.4 \quad 1$$

$$v_1 = 0.1$$

$$v_2 = 0.2$$

$$v_3 = -0.3$$

$$v_4 = -0.1$$

$$v_5 = 0.4$$

$$\boxed{Z = n_1 w_1 + n_2 w_2 + b}$$

$$Z_1 = 4(0.1) + 2(0.2) + 1 = 1$$

$$Z_2 = 4(0.3) + 2(0.1) + 1 = 2$$

$$Z_3 = 4(0.2) + 2(0.4) + 1 = 2.6$$

$$Z_4 = 4(0.1) + 2(0.3) + 1 = 2$$

$$Z_5 = 4(-0.2) + 2(0.4) + 1 = 1$$

Hidden Neurons:

$$h_1 = \sigma(z_1) = \frac{1}{1 + e^{-z_1}} = 0.731$$

$$h_2 = \sigma(z_2) = 0.880$$

$$h_3 = \sigma(z_3) = 0.930$$

$$h_4 = \sigma(z_4) = 0.880$$

$$h_5 = \sigma(z_5) = 0.731$$

$$z_{out} = h_1 v_1 + h_2 v_2 + h_3 v_3 + h_4 v_4 + h_5 v_5$$

$$= 0.73 \times 0.1 + 0.88 \times 0.2 + 0.93 \times (-0.3) + 0.88 \times -0.1 + 0.73 \times 0.4$$

$$\boxed{z_{out} = 0.174}$$

$$\hat{y} = \sigma(z_{out}) = \frac{1}{1 + e^{-0.174}} = 0.543$$

$$\hat{y} \approx 0.5 \Rightarrow 1 \quad y = 1$$

$$\boxed{y = 1}$$

Sample 2 (1, 0, 1)

$$z_1 = 1(0.1) + 0(-0.2) + 1 \\ \Rightarrow 1.1$$

$$z_2 = 1(0.3) + 0(-0.1) + 1 \\ \Rightarrow 1.3$$

$$z_3 = 1(0.2) + 0(0.4) + 1 \\ \Rightarrow 1.2$$

$$z_4 = 1(0.1) + 0(0.3) + 1 \Rightarrow 1.1$$

$$z_5 = 1(-0.2) + 0(0.4) + 1 \Rightarrow 0.8$$

Hidden neurons.

$$h_1 = 0.750$$

$$h_2 = 0.785$$

$$h_3 = 0.768$$

$$h_4 = 0.750$$

$$h_5 = 0.689$$

$$z_{out} = 0.750(0.2) + 0.785(0.2) + 0.768(-0.3) + 0.750(-0.1) + 0.689(0.4)$$

$$z_{out} = 0.2022$$

$$\hat{y} = 0.550 \quad y = 1, \quad y \neq \hat{y}$$

updating weights

$$y = 0, \quad \hat{y} = 0.55, \quad \eta = 0.1$$

$$\eta(y - \hat{y}) = 0.1(0 - 0.55) = -0.055$$

$$V_i^{new} = V_i^{old} + \eta(y - \hat{y})h_i$$

$$V_1^{new} = 0.1 + (-0.055)(0.750) = 0.05875$$

$$V_2^{new} = 0.2 + (-0.055)(0.785) = 0.1568$$

$$V_3^{new} = -0.3 + (-0.055)(0.768) = -0.34224$$

$$V_4^{new} = -0.1 + (-0.055)(0.750) = -0.14125$$

$$V_5^{new} = 0.4 + (-0.055)(0.689) = 0.362105$$

Sample 3. (6, 3, 1)

$$z_1 = 6(0.1) + 3(-0.2) + 1 = 1$$

$$z_2 = 6(0.3) + 3(-0.1) + 1 = 2.5$$

$$z_3 = 6(0.2) + 3(0.4) + 1 = 3.4$$

$$z_4 = 6(0.1) + 3(0.3) + 1 = 2.5$$

$$z_5 = 6(-0.2) + 3(0.4) + 1 = 1$$

Hidden neurons:

$$h_1 = 0.731$$

$$h_2 = 0.924$$

$$h_3 = 0.967$$

$$h_4 = 0.924$$

$$h_5 = 0.731$$

$$z_{out} = 0.731(0.1) + 0.924(0.2) + 0.967(-0.3) + 0.924(-0.1) + 0.731(0.4)$$

$$\boxed{z_{out} = 0.1678}$$

$$\hat{y} = 0.54 \cdot \hat{y} = 1 \quad y = 1$$

$$\boxed{y = \hat{y}}$$

$$z_1 = 2(0.1) + 1(-0.2) + 1 = 1$$

$$z_2 = 2(0.3) + 1(-0.1) + 1 = 1.5$$

$$z_3 = 2(0.2) + 1(0.4) + 1 = 1.8$$

$$z_4 = 2(0.5) + 1(0.3) + 1 = 1.5$$

$$z_5 = 2(-0.2) + 1(0.4) + 1 = 1$$

16. Alex Haulton:

$$w_1 = 0.731$$

$$w_2 = 0.817$$

$$w_3 = 0.858$$

$$w_4 = 0.817$$

$$w_5 = 0.721$$

$$Z_{out} = 0.731(0.1) + 0.817(0.2) + 0.858(0.3) \\ + 0.817(0.1) + 0.721(0.4)$$

$$Z_{out} = 0.1873$$

$$\hat{y} = 0.547 \Rightarrow \hat{y} = 1 \quad y_{\text{target}} = 1$$

Update w_i :

$$y = 0 \quad \hat{y} = 0.547 \quad \eta = 0.1$$

$$\eta(y - \hat{y}) = 1.1(0 - 0.547) = -0.0547$$

$$w_1 = 0.1 + (-0.0547)(0.731) = 0.060$$

$$w_2 = 0.2 + (-0.0547)(0.817) = 0.155$$

$$w_3 = 0.3 + (-0.0547)(0.858) = 0.346$$

$$w_4 = 0.1 + (-0.0547)(0.817) = 0.144$$

$$w_5 = 0.4 + (-0.0547)(0.721) = 0.360$$